holibob

The Holibob API - Product Discovery

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1 Introduction

The **Product Discovery Query** is an **AI-powered recommendation** endpoint which delivers intelligent, contextually relevant recommendations based on traveller trip information. This endpoint is designed to power entire user interfaces by recommending not only **products** but filters, search terms and more.

This endpoint is an **alternative** to our **Product List Query,** which is a more traditional product endpoint with complex filtering, sorting and pagination.

2 Design Principles

Product Discovery is different from traditional product list endpoints and follows these principles.

Principle	
Free Text vs Data	Product Discovery supports free text input, allowing queries to be written entirely in natural language, just like a human conversation. This flexibility makes the endpoint usable with many different applications, including use with Large Language Models (LLMs), enabling integration into AI-powered assistants or conversational UIs. For every free text field, you also have the option of passing the data directly. You cannot pass both.
Personalisation	Every element of the response is personalised based on the contextual data provided in the request.
Ordering	Product Discovery automatically orders all recommendations by their likelihood of appealing to the user. The order cannot be controlled explicitly by an input. Here are some examples of data our models use: ▪ Past product performance for this type of trip ▪ Product's distance from the user's accommodation ▪ Weather during travel dates ▪ Seasonal trends within the destination ▪ Real-time user clickstream data ▪ And more…
Pagination	Traditional pagination is not supported. This is because the model that orders products is continuously changing (for example, real-time user clickstream data will impact orders).

	Instead, clients can provide a list of **seen product IDs** to ensure that subsequent requests return new, unrepeated recommendations. We suggest creating an infinite-scroll or "load more" experience.
Selected Destination	All Product Discovery responses are generated **within the context of a destination**, and all products will be relevant to that destination. You can pass the destination in the inputs, or the system will select one for you. The API can also **suggest alternative destinations** that may also be of interest, and you can switch destinations using the inputs.

3 Inputs

Product Discovery takes the following inputs. Each input has an option to pass **free text** OR pass **data**.

```
{
  where: {...},
  when: {...},
  who: {...},
  what: {...},
  consumerTripSelector: {...}
}
```

Input Variable	Description	Schema	Example/s

where	Pass information about **where** the traveller is starting from. This can be an airport, a hotel, or a destination.
	This is **required** to determine the **destination** and the **recommended destination list**.
		`where: { freeText: String, data: { destinationId } }`	`where: { freeText: "Paris, France" }`
			`where: { freeText: "London Gatwick Airport" }`
			`where: {`

			`data: { destinationId: "destination-id-001" } }`

when	Pass information about when the traveller is looking for things to do.	`when: { freeText: String, data: { startDate: DateTime, endDate: DateTime, duration: { min: Number, max: Number, unit: MINUTE | HOUR | DAY } } }`	`when: { freeText: "Christmas Holidays" }`
			`when: { freeText: "Tonight" }`
			`when: { data: { startDate: "2025-03-01T00:00:00.000Z", endDate: "2025-03-08T00:00:00.000Z", duration: {max: 90, unit: MINUTE} } }`

who	This is used to pass any information about who the traveller/s is.	`who: { freeText: String, data: { personList: [{ age: Number }], interestList: [String] } }`	`who: { freeText: "Two adults and a young child interested in history" }` `who: {`

			`freeText: "A stag do of 10"`
			`}`
			
			`who: {`
			`data: {`
			`personList: [{ age: 20 }, { age: 22 }, { age: 3 }],`
			`interestList: ["History"]`
			`}`
			`}`

what	This is used to pass any information about what the traveller is looking for.	`what: {`
		`freeText: String,`
		`data: {`
		`tagIdList: [String],`
		`searchTerm: String,`
		`price: { min: Number, max: Number },`
		`rating: { min: Number, max: Number }`
		`}`
		`}`
			`what: {`
			`freeText: "The most booked cooking experiences which are less than 2 hours long and under £30"`
			`}`
			
			`what: {`
			`data: {`
			`tagIdList: ["tag-id-001"],`
			`searchTerm: "Cooking experiences",`
			`price: { max: 30 },`
			`rating: { min: 4 }`
			`}`
			`}`

consumerTripSelector	This is used when we already have consumer trip data in our system and can use that to personalise the response.	`consumerTripSelector: {`
		`type: REFERENCE | ID`
		`value: String`
			`consumerTripSelector: {`
			`type: REFERENCE`

		`}`	`value: "partner-id-001"`
			`}`
			
			`consumerTripSelector: {`
			`type: ID`
			`value: "holibob-id-001"`
			`}`

4 Output

```
{
  selectedDestination: {...},
  recommendedDestinationList: {...},
  recommendedTagList: {...},
  recommendedSearchTermList: {...},
  recommendedProductList(seenProductIdList: []): {...}
}
```

Output Variable	Description	Schema

selectedDestination	The selected destination.	`selectedDestination: {`
		`id: String,`
		`name: String,`
		`...`
		`}`

recommendedDestination List	Other destinations which are most relevant to the trip.	`recommendedDestinationList: {`
		`nodes: {`
		`id: String,`
		`name: String,`
		`weatherForecast: {`
		`date: Date,`

		`type: CLEAR_DAY | CLOUDY | FOG ...,`
		`temperature: {`
		`value: Number,`
		`unit: CELSIUS | FAHRENHEIT`
		`}`
		`}`
		`...`
		`}`
		`}`

recommendedTagList	Tags which are most relevant to the trip and can be used for filtering.
	e.g Most Likely to Sell Out
		`recommendedTagList: {`
		`nodes: {`
		`id: String,`
		`name: String,`
		`...`
		`}`
		`}`

recommendedSearchTermList	Search terms which are most relevant to the trip.
	e.g Covent Garden Food Tours
		`recommendedSearchTermList(count: number): {`
		`nodes: {`
		`searchTerm: String,`
		`...`
		`}`
		`}`

recommendedProductList	Products which the traveller is most likely to purchase.	`recommendedProductList(`
		`seenProductIdList: ["product-id-001"],`
		`count: number`
		`): {`
		`nodes: {`
		`id: String,`
		`name: String,`
		`...`
		`}`

```
}
```

5 Examples

5.1 1. Recommended Destinations

You can use this field to get recommended destinations based on trip information. In the `where` field, you can pass a destination or something more precise, like an airport or hotel address.

Request

```
query Example {
  productDiscovery(where: { freeText: "London Gatwick Airport" }) {
    recommendedDestinationList: {
      nodes {
        id
        name
      }
    }
  }
}
```

Response

```
{
  recommendedDestinationList: {
    nodes: [
      {
        id: "destination-id-001",
        name: "London"
      },
      {
        id: "destination-id-002",
        name: "Cambridge"
      },
      {
        id: "destination-id-003",
        name: "Oxford"
```

```
      }
    ]
  }
}
```

5.2 2. Recommended Search Terms

You can use this field to get recommended search terms based on trip information. This can then be passed into the `what.data.searchTerm` field.

Request

```
query Example {
  productDiscovery(
    where: { freeText: "London, England" },
    who: { freeText: "Young Family" },
    when: {
      data: {
        date: { start: "2025-03-01T00:00:00.000Z", end: "2025-03-08T00:00:00.000Z" }
      }
    }
  )
  {
    recommendedSearchTermList: {
      nodes {
        id
        name
      }
    }
  }
}
```

Response

```
{
  recommendedSearchTermList: {
    nodes: [
      { searchTerm: "London Eye" },
      { searchTerm: "Tower of London" },
      { searchTerm: "Harry Potter Experiences" },
```

```
      ...
    ]
  }
}
```

5.3 3. Recommended Products

You can ask for a list of recommended products for a trip.

Request

```
query Example {
  productDiscovery(
    where: { data: { destinationId: "destination-id-001" } },
    who: { freeText: "Young Family interested in History" },
    when: {
      data: {
        date: { start: "2025-03-01T00:00:00.000Z", end: "2025-03-08T00:00:00.000Z" }
      }
    }
  )
  {
    recommendedProductList: {
      nodes {
        id
        name
      }
    }
  }
}
```

Response

```
{
  recommendedProductList: {
    nodes: [
      {
        id: "product-id-001",
        ...
      },
      {
```

```
        id: "product-id-002",
        ...
      },
      {
        id: "product-id-003",
        ...
      }
    ]
  }
}
```